

```
users@Lenovo:~/myProject$ python3 c_import pandas  
et'); print(df.head(50).to_string())"
```

temperature_c	humidity_pct	co2_ppm	yield_kg
22.46	91.4	936.0	18.09
20.44	84.8	905.0	17.35
23.13	84.5	944.0	17.52
23.41	87.6	885.0	18.12
19.07	89.5	922.0	16.98
20.05	87.0	913.0	17.11
22.19	91.0	962.0	17.59
21.53	89.6	965.0	17.31
21.97	89.5	770.0	17.79
20.72	88.7	720.0	16.66
23.32	94.0	980.0	18.08
23.17	86.4	995.0	16.60
22.10	81.0	818.0	16.69
23.69	91.8	751.0	18.85
22.70	85.6	908.0	17.70
20.71	87.3	974.0	17.37
22.55	90.9	1044.0	17.09
20.56	82.2	941.0	15.82
23.32	83.2	870.0	17.30
21.93	82.2	829.0	17.08
21.72	84.6	901.0	16.99
20.98	88.3	876.0	17.52
23.83	88.6	819.0	17.76
21.77	87.8	1064.0	16.97
21.36	82.8	1043.0	15.97
21.47	80.1	991.0	16.69
22.80	87.2	826.0	17.59
22.55	85.6	968.0	16.73
22.62	88.4	951.0	17.18
22.65	89.1	935.0	17.29
25.21	87.4	1000.0	17.80
21.39	89.3	951.0	17.87
21.23	87.7	959.0	16.79
20.78	88.6	951.0	16.64
22.92	84.9	927.0	16.22
23.69	86.5	757.0	17.47
21.83	87.6	907.0	16.98
20.74	89.5	856.0	16.64
20.76	85.8	798.0	17.62
22.98	88.6	1035.0	17.17
23.11	86.2	1038.0	17.81
22.81	86.6	1009.0	17.19
21.00	89.5	920.0	16.99
22.35	81.0	1008.0	16.70
22.18	83.1	901.0	17.01
22.33	82.6	916.0	17.89
23.31	80.0	813.0	16.21
22.34	85.0	932.0	16.08
23.02	89.2	905.0	17.29
22.10	86.1	796.0	17.39

```
Dataset Shape:  
(365, 5)
```

```
Data Types:  
timestamp          str  
temperature_c      float64  
humidity_pct       float64  
co2_ppm            float64  
yield_kg           float64  
dtype: object
```

```
First 5 Rows:
```

```
Missing Values:
timestamp      0
temperature_c  0
humidity_pct   0
co2_ppm        0
yield_kg       0
```

```
(myenv) PS C:\Users\Lenovo\MyProject> python -c "import pandas as pd; df=pd.read_csv('data/raw/polyhouse_sensors.csv'); df['timestamp']=pd.to_datetime(df['timestamp']); df.to_parquet('data/processed/02_cleaned.parquet',index=False); print('Cleaned dataset saved successfully.')"
Cleaned dataset saved successfully.
```

```
(myenv) PS C:\Users\Lenovo\MyProject> python -c "import pandas as pd; df=pd.read_csv('data/raw/polyhouse_sensors.csv'); print('Duplicate Rows:'); print(df.duplicated().sum())"
Duplicate Rows:
```